

PORTABLE ULTRASONIC TRENCH LOGGER For Optical Fibre Cable

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In modern telecom infrastructure deployments, optical fibre cable (OFC) is typically laid in underground trenches, where maintaining the specified trench depth is critical for safety, long-term reliability, and protection against accidental damage. Traditionally, excavation is carried out manually or using trenching machines, while trench depth is measured with tapes or scales and the running length is recorded separately. This conventional method is labour-intensive, time-consuming, and susceptible to human error.

To address these limitations, a portable ultrasonic trench depth and location logger system is proposed. It provides an automated, accurate, and user-friendly solution for real-time field measurements. The system uses an ultrasonic depth sensor, such as the HC-SR04, mounted on a handheld frame to measure trench depth without physical contact. In addition, a GPS module captures precise latitude and longitude coordinates, enabling the geotagging of every measurement.

At the core of the system, an Arduino Uno synchronises depth measurements with GPS data and logs them onto an SD card in CSV format for easy analysis and visualisation. The recorded data can also be converted into KML format for use in mapping applications such as Google Earth. Fig. 1 shows the author's prototype.

For enhanced usability, a push-button switch allows the operator to initiate data logging, while a buzzer provides audible feedback each time a reading is successfully recorded. This improves operational reliability

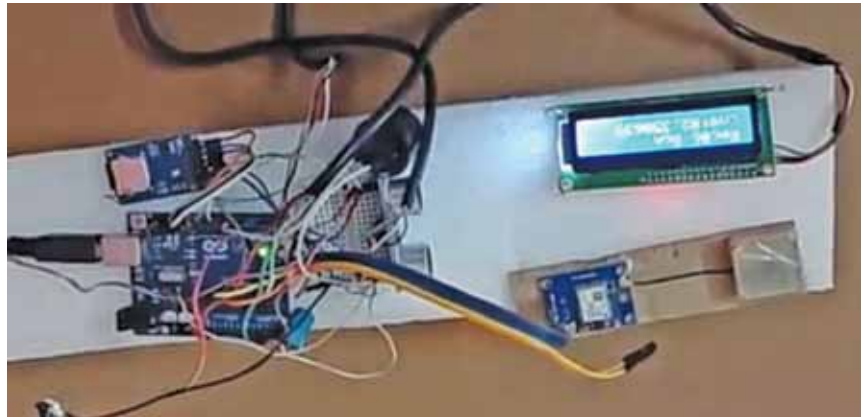


Fig. 1: Author's prototype

BILL OF MATERIALS

Components	Description/Function	Quantity
Arduino Uno R3	The main microcontroller board that controls all peripherals	1
HC-SR04 ultrasonic sensor	Measures trench depth (distance to bottom)	1
NEO-6M GPS module	Provides latitude and longitude for location tagging	1
I ² C 16×2 LCD display	Displays distance, GPS data, and record number	1
Micro SD card module (SPI)	Logs data (distance, GPS) to SD card	1
Push button (tactile switch)	Triggers manual data logging when pressed	1
Active buzzer (5V)	Provides audio feedback when data is logged	1
Power bank (5V USB output)	Portable power supply for field use	1
Connecting wires (male-male, male-female)	For all interconnections between modules	1
Breadboard/PCB mount	For prototyping or mounting modules	1
Mounting setup/handle frame	Ultrasonic sensor facing downward into the trench	1

and field usability. The logged data can be further processed to generate depth profiles, reports, and geospatial maps, enabling efficient monitoring and documentation of OFC trench-laying operations.

The components required to build this system are listed in Bill of Materials table.

Circuit and working

Fig. 2 shows the circuit diagram of the portable ultrasonic trench logger system designed for optical fibre cable (OFC) applications. The

system is built around an Arduino Uno (MOD2), interfaced with a 16 × 2 LCD (MOD1), an SD card module (MOD3), a Neo-6M GPS module (MOD4), and an HC-SR04 ultrasonic sensor (S1), along with supporting components.

The Arduino Uno serves as the central controller, coordinating all sensing, processing and data-logging operations. The ultrasonic sensor is connected to digital pins 7 (Trig) and 8 (Echo) to measure trench depth using the time-of-flight principle, enabling accurate non-contact